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Shielded electrically conductive cable connector

Abstract

A shielded electrically conductive path is provided with a shield terminal including an inner conductor and an outer conductor surrounding the inner conductor, a shielded cable including a core wire fixed to a rear end part of the inner conductor, a braided wire surrounding the core wire and a sheath surrounding the braided wire, and a sleeve mounted on an outer periphery of a front end part of the sheath. In a front end part of the shielded cable, the braided wire extending from a front end of the sheath is folded rearward and put on an outer periphery of the sleeve and a crimping portion in a rear end part of the outer conductor is crimped to the sleeve while surrounding an outer periphery of a folded portion of the braided wire.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase of PCT application No. PCT/JP2021/016420, filed on 23 Apr. 2021, which claims priority from Japanese patent application No. 2020-085129, filed on 14 May 2020, all of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a shielded electrically conductive path.

BACKGROUND

(3) Patent Document 1 discloses a structure for connecting such a terminal fitting that an inner conductor terminal is surrounded with an outer conductor terminal and a shielded cable. The shielded cable includes a core wire, a braided wire surrounding the core wire and an outer sheath surrounding the braided wire. In a front end part of the shielded cable, a ring member is externally fit to the outer periphery of the outer sheath and a front end part of the braided wire is folded rearward and put on the outer periphery of the ring member. A barrel portion in a rear end part of the outer conductor terminal is crimped to the ring member while being held in close contact with the outer periphery of the braided wire and the crimped ring member is held in close contact with the outer periphery of the outer sheath, whereby the outer conductor terminal is fixed to the front end part of the shielded cable.

PRIOR ART DOCUMENT

Patent Document

(4) Patent Document 1: JP 2015-226383 A

SUMMARY OF THE INVENTION

Problems to be Solved

(5) In recent years, a higher transmission speed is required in transmission circuits of automotive vehicles. To improve and stabilize communication performance in high-speed transmission, it is necessary to keep an interval between a core wire and a braided wire constant and achieve impedance matching by preventing the deformation of a shielded cable. If a barrel portion of an outer conductor terminal is strongly crimped to a ring member, the shielded cable is deformed to be squeezed in a radial direction. To avoid a reduction in communication performance, a crimping force of the barrel portion needs to be suppressed.

(6) However, if the crimping force of the barrel portion is weakened, a slip occurs between the outer peripheral surface of the outer sheath and the inner peripheral surface of the ring member and the shielded cable is relatively displaced rearward with respect to the outer conductor terminal and the inner conductor terminal when the shielded cable is pulled rearward. Thus, a load concentrates on a connected part of the inner conductor terminal and the core wire, whereby the core wire may come out from the inner conductor terminal or may be broken. Therefore, communication performance is reduced if the connection reliability of the terminal fitting and the shielded cable is prioritized.

(7) A shielded electrically conductive path of the present disclosure was completed on the basis of the above situation and aims to prevent a reduction in communication performance.

Means to Solve the Problem

(8) The present disclosure is directed to a shielded electrically conductive path with a shield terminal including an inner conductor and an outer conductor surrounding the inner conductor, a shielded cable including a core wire fixed to a rear end part of the inner conductor, a braided wire surrounding the core wire and a sheath surrounding the braided wire, and a sleeve mounted on an outer periphery of a front end part of the sheath, the braided wire extending from a front end of the sheath being folded rearward and put on an outer periphery of the sleeve and a crimping portion in

a rear end part of the outer conductor being crimped to the sleeve while surrounding an outer periphery of a folded portion of the braided wire in a front end part of the shielded cable, and a front end of the sleeve being located at the same position as or forward of the front end of the sheath in a front-rear direction.

Effect of the Invention

(9) According to the present disclosure, it is possible to prevent a reduction in communication performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a shielded electrically conductive path of a first embodiment.
- (2) FIG. 2 is a section of the shielded electrically conductive path.
- (3) FIG. 3 is a perspective view of an upper shell.
- (4) FIG. 4 is a perspective view of a lower shell.
- (5) FIG. 5 is a perspective view of a sleeve in a state before being mounted on a shielded cable.
- (6) FIG. 6 is a plan view showing a state where the sleeve is externally fit to the shielded cable.
- (7) FIG. 7 is a plan view showing a state where a sleeve is externally fit to a shielded cable in a second embodiment.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

Description of Embodiments of Present Disclosure

- (8) First, embodiments of the present disclosure are listed and described.
- (9) (1) The shielded electrically conductive path of the present disclosure is provided with a shield terminal including an inner conductor and an outer conductor surrounding the inner conductor, a shielded cable including a core wire fixed to a rear end part of the inner conductor, a braided wire surrounding the core wire and a sheath surrounding the braided wire, and a sleeve mounted on an outer periphery of a front end part of the sheath, the braided wire extending from a front end of the sheath being folded rearward and put on an outer periphery of the sleeve and a crimping portion in a rear end part of the outer conductor being crimped to the sleeve while surrounding an outer periphery of a folded portion of the braided wire in a front end part of the shielded cable, and a front end of the sleeve being located at the same position as or forward of the front end of the sheath in a front-rear direction.
- (10) According to the configuration of the present disclosure, if a crimping force by the crimping portion is reduced to avoid a reduction in communication performance, the shielded cable is relatively displaced rearward with respect to the outer conductor when a rearward pulling force is applied to the shielded cable. However, at the time of or immediately after the start of a relative displacement of the shielded cable, a bent portion on the front end of the folded portion of the braided wire is caught by the front end of the sleeve. Thus, a rearward relative displacement of the shielded cable is hindered. Therefore, a reduction in communication performance can be prevented while a connected state of the shield terminal and the shielded cable is maintained.
- (11) (2) Preferably, an expanded diameter portion having a diameter increasing toward front is formed in a front end part of the sleeve. According to this configuration, even if the sleeve is relatively displaced rearward with respect to the crimping portion when a rearward pulling force is applied to the shielded cable, an outer periphery of the expanded diameter portion is caught by the crimping portion, whereby a rearward relative displacement of the sleeve with respect to the crimping portion is prevented.
- (12) (3) Preferably, in (2), a rear end of the expanded diameter portion is located rearward of the front end of the sheath. According to this configuration, when a rearward pulling force is applied to the shielded cable, the front end part of the sheath is caught by the inner peripheral surface of the

expanded diameter portion, whereby a rearward relative displacement of the shielded cable can be prevented.

(13) (4) Preferably, the crimping portion is formed with a biting portion configured to come into contact with and bite into the sheath. According to this configuration, a rearward relative displacement of the shielded cable with respect to the shield terminal can be prevented by the biting action of the biting portion.

(14) (5) Preferably, the front end of the sleeve is located forward of the front end of the sheath, and the sleeve is formed with a cut portion for exposing the front end of the sheath to an outer peripheral surface side of the sleeve. According to this configuration, a positional relationship of the front end of the sleeve and the front end of the sheath can be confirmed through the cut portion.

Details of Embodiments of Present Disclosure

First Embodiment

(15) A first specific embodiment of a shielded electrically conductive path A of the present disclosure is described below with reference to FIGS. **1** to **6**. Note that the present invention is not limited to these illustrations and is intended to be represented by claims and include all changes in the scope of claims and in the meaning and scope of equivalents. In this embodiment, an oblique left lower side in FIGS. **1**, **3** and **4** and a left side in FIGS. **2** and **6** are defined as a front side concerning a front-rear direction. Upper and lower sides shown in FIGS. **1**, **3** and **4** are directly defined as upper and lower sides concerning a vertical direction.

(16) The shielded electrically conductive path A of the first embodiment includes a shielded cable **10**, a sleeve **20** externally fit to the shielded cable **10** and a shield terminal **25** connected to a front end part of the shielded cable **10** using the sleeve **20**.

(17) The shielded cable **10** is such that a plurality of coated wires **11** are embedded in an insulating member (not shown) having a circular cross-section, the outer periphery of the insulating member is surrounded with a braided wire **15** having a shielding function, and the braided wire **15** is surrounded with a hollow cylindrical sheath **18**. As shown in FIG. **2**, the sheath **18** and the insulating member are removed in a front end part of the shielded cable **10**, and the plurality of coated wires **11** are exposed forward of the insulating member in an individually bendable state. In a front end part of each coated wire **11**, an insulation coating **13** is removed to expose a front end part of an electrically conductive core wire **12**. The braided wire **15** is a flexible tubular member formed by braiding strands made of metal. A front end part of the braided wire **15** is exposed while extending further forward than a front end **18F** of the sheath **18**.

(18) The electrically conductive sleeve **20** made of metal or the like is externally fit to the outer periphery of a front end part of the sheath **18**. As shown in FIG. **5**, the sleeve **20** is formed of a plate-like member having a constant width. The sleeve **20** is bent to have a hollow cylindrical shape with an axis oriented in the front-rear direction, and is externally fit to the front end part of the sheath **18** to surround the sheath **18** over an entire periphery. A front end **20F** of the sleeve **20** is located forward of the front end **18F** of the sheath **18**. That is, a front end part of the sleeve **20** projects further forward than the front end **18F** of the sheath **18**.

(19) The sleeve **20** bent into a hollow cylindrical shape includes a constant diameter portion **21** and an enlarged diameter portion **22**. The constant diameter portion **21** has a hollow cylindrical shape having constant inner and outer diameters over an entire length in an axial direction. The expanded diameter portion **22** extends forward from the front end of the constant diameter portion **21**. The expanded diameter portion **22** has such a tapered shape, i.e. a truncated conical shape that inner and outer diameters gradually increase toward the front. In the front-rear direction (axial direction of the shielded cable **10**), the constant diameter portion **21** is entirely arranged in a region behind the front end **18F** of the sheath **18**. A rear end **22R** of the expanded diameter portion **22** is located rearward of the front end **18F** of the sheath **18**, and a front end **22F** of the expanded diameter portion **22** (i.e. front end **20F** of the sleeve **20**) is located forward of the front end **18F** of the sheath **18**.

(20) As shown in FIG. 2, in a region behind an exposed region of the core wire **12**, out of the front end part of the shielded cable **10**, the front end part of the braided wire **15** is bent to turn rearward and toward an outer peripheral side near the front end **18F** of the sheath **18**. Out of the braided wire **15**, a folded portion **17** rearward of a bent portion **16** covers the outer periphery of the sleeve **20**. The bent portion **16** of the braided wire **15** is arranged at a position where the bent portion **16** is in contact with and caught by the front end **20F** of the sleeve **20** or at a position where the bent portion **16** is proximately facing the front end **20F** of the sleeve **20** from front.

(21) The shield terminal **25** includes inner conductors **26** individually connected to front end parts of the core wires **12** of the respective coated wires **11**, a dielectric **27** accommodating a plurality of the inner conductors **26** and an outer conductor **28** mounted on the dielectric **27** while surrounding the outer periphery of the dielectric **27**. The outer conductor **28** functionally includes a shell body portion **29** in the form of a rectangular tube constituting a front end part of the outer conductor **28** and a hollow cylindrical crimping portion **30**. The crimping portion **30** is connected to the rear end of the shell body portion **29** and constitutes a rear end part of the outer conductor **28**. In terms of component configuration, the outer conductor **28** is configured by vertically uniting an upper shell **31** formed by applying bending and the like to a metal plate material and a lower shell **32** formed by applying bending and the like to a metal plate material.

(22) As shown in FIG. 3, the upper shell **31** is a single component including an upper body portion **33** in the form of a box with an open lower surface, a base plate portion **34**, a first crimp portion **35** and a second crimp portion **36**. The base plate portion **34**, the first crimp portion **35** and the second crimp portion **36** constitute the crimping portion **30**. The base plate portion **34** extends rearward from the rear end edge of the upper body portion **33** and has an arched shape curved to bulge upward.

(23) In a state where the crimping portion **30** (outer conductor **28**) is not crimped to the shielded cable **10**, the first crimp portion **35** extends obliquely to a right lower side along a circumferential direction from a right side edge part of the base plate portion **34**. A pair of first locking portions **37** spaced apart in the front-rear direction are formed on an extending end part of the first crimp portion **35**. In the state where the crimping portion **30** (outer conductor **28**) is not crimped to the shielded cable **10**, the second crimp portion **36** extends obliquely to a left lower side along the circumferential direction from a left side edge part of the base plate portion **34**. A second locking portion **38** is formed on an extending end part of the second crimp portion **36**.

(24) A plurality of (four in the first embodiment) biting portions **39** are formed on the rear end edge of the crimping portion **30**. The biting portion **39** is in the form of a projection formed by bending a part of a rear end edge part of the crimping portion **30** inwardly (in a direction toward the outer peripheral surface of the shielded cable **10**). A pair of left and right biting portions **39** are formed on the rear end edge of the base plate portion **34**, and one biting portion **39** is formed on each of the rear end edge of the first crimp portion **35** and the rear end edge of the second crimp portion **36**.

(25) The lower shell **32** is a single component including a lower body portion **40** in the form of a box with an open upper surface and a restricting portion **41**. The restricting portion **41** is cantilevered rearward from the rear end edge of the lower body portion **40** and constitutes the crimping portion **30**.

(26) Next, the assembling of the shielded electrically conductive path A is described. If the upper shell **31** and the lower shell **32** are assembled to be vertically united, the upper body portion **33** and the lower body portion **40** are vertically united to sandwich the dielectric **27**, whereby the shell body portion **29** in the form of a rectangular tube is configured. The entire dielectric **27** and the exposed regions of the coated wires **11** are accommodated in the shell body portion **29**. In the state where the outer conductor **28** and the shielded cable **10** are not crimped, the base plate portion **34** of the crimping portion **30** and the restricting portion **41** are located to vertically sandwich the front end part of the shielded cable **10**.

(27) In this state, crimping is performed with the crimping portion **30** and the front end part of the

shielded cable **10** set in an applicator (not shown). In a crimping step, the first and second crimp portions **35**, **36** are deformed to be pressed against the outer periphery of the folded portion **17** of the braided wire **15** and the first and second locking portions **37**, **38** are locked to the restricting portion **41** from opposite sides in the circumferential direction. By the locking action of the first and second locking portions **37**, **38**, the crimping portion **30** is crimped to the outer peripheries of the folded portion **17** of the braided wire **15** and the sleeve **20** and crimped to the shielded cable **10**.

(28) In a crimped state, at the same time as the folded portion **17** of the braided wire **15** is sandwiched between the inner peripheral surface of the crimping portion **30** and the outer peripheral surface of the sleeve **20** and the outer conductor **28** and the braided wire **15** are conductively connected, the outer conductor **28** and the shield terminal **25** are fixed to the shielded cable **10** with relative displacements in the front-rear direction restricted. Further, the biting portions **39** bite into the outer periphery of the sheath **18** at positions behind and very close to the sleeve **20**. By this biting of the biting portions **39**, relative displacements of the shielded cable **10** with respect to the crimping portion **30** in the front-rear direction and circumferential direction are prevented. In the above way, the connection of the shield terminal **25** and the shielded electrically conductive path A is completed at the same time as the assembling of the outer conductor **28** is completed.

(29) If the crimping portion **30** of the outer conductor **28** is crimped to the front end part of the shielded cable **10**, the sleeve **20** is slightly reduced in diameter and deformed by the crimping portion **30**. Thus, a tapered portion **42** in conformity with the expanded diameter portion **22** of the sleeve **20** is formed in a part of the crimping portion **30** forward of the first and second crimp portions **35**, **36**. Further, by being slightly reduced in diameter and deformed by the crimping portion **30**, the sleeve **20** comes into surface contact with an outer peripheral part of the sheath **18** such that the sheath **18** is reduced in diameter and deformed. A deformation amount to reduce the diameter of the sheath **18** by the sleeve **20** is such an amount as not to affect impedance matching. Out of the outer peripheral part of the sheath **18**, a front end region reduced in diameter and deformed by the expanded diameter portion **22** has a tapered shape having an outer diameter increasing toward the front, similarly to the tapered shape of the inner peripheral surface of the expanded diameter portion **22**.

(30) If a crimping force by the crimping portion **30** is strengthened, the deformation amount to reduce the diameter of the sheath **18** by the sleeve **20** increases and the sleeve **20** and the sheath **18** are hardly relatively displaced in the front-rear direction. Thus, a fixing force of the crimping portion **30** to the shielded cable **10**, i.e. a holding force by the crimping portion **30**, increases. However, on the other hand, a radial distance between the core wire **12** and the braided wire **15** becomes largely different between a part of the sheath **18** reduced in diameter and deformed by the sleeve **20** and a part thereof not reduced in diameter and deformed. Thus, impedance mismatching occurs and communication performance is reduced.

(31) Since the crimping force by the crimping portion **30** is suppressed to avoid a reduction in communication performance in the shielded electrically conductive path A of the first embodiment, there is a concern for a reduction in fixing force between the shield terminal **25** and the shielded cable **10**. As a countermeasure against that, the expanded diameter portion **22** is formed in the front end part of the sleeve **20** and the front end part of the sheath **18** is held in close contact with the inner peripheral surface of the expanded diameter portion **22**. Since the front end part of the sheath **18** comes into contact with and is caught by the inner peripheral surface of the tapered expanded diameter portion **22** from front when the shielded cable **10** is pulled rearward, a rearward relative displacement of the shielded cable **10** with respect to the shield terminal **25** can be prevented.

(32) Further, the biting portions **39** formed on the crimping portion **30** are caused to bite into the outer peripheral part of the sheath **18** as a countermeasure in the case where a rearward pulling force applied to the shielded cable **10** exceeds a hooking force of the sheath **18** to the expanded diameter portion **22**. When the shielded cable **10** is pulled rearward, the sheath **18** is caught by the

biting portions 39, whereby a rearward relative displacement of the shielded cable 10 with respect to the shield terminal 25 can be prevented.

(33) Further, the front end 20F of the sleeve 20 is arranged forward of the front end 18F of the sheath 18 and the bent portion 16 of the braided wire 15 is in contact with or proximately facing the front end 20F of the sleeve 20 as a countermeasure in the case where a rearward pulling force applied to the shielded cable 10 exceeds a locking force by the biting portions 39. According to such a positional relationship, if a rearward pulling force is applied to the shielded cable 10, the bent portion 16 of the braided wire 15 comes into contact with the front end 20F of the sleeve 20 from front. Since the folded portion 17 of the braided wire 15 is sandwiched between the sleeve 20 and the crimping portion 30, a rearward relative displacement of the shielded cable 10 can be reliably prevented by the bent portion 16 being caught by the front end 20F of the sleeve 20.

(34) The shielded electrically conductive path A of the present disclosure is provided with the shield terminal 25, the shielded cable 10 and the sleeve 20. The shield terminal 25 includes the inner conductors 26 and the outer conductor 28 surrounding the inner conductors 26. The shielded cable 10 includes the core wires 12 fixed to the rear end parts of the inner conductors 26, the braided wire 15 surrounding the core wires 12 and the sheath 18 surrounding the braided wire 15. The sleeve 20 is mounted on the outer periphery of the front end part of the sheath 18. In the front end part of the shielded cable 10, the braided wire 15 extending from the front end 18F of the sheath 18 is folded rearward and put on the outer periphery of the sleeve 20. The crimping portion 30 in the rear end part of the outer conductor 28 is crimped to the sleeve 20 while surrounding the outer periphery of the folded portion 17 of the braided wire 15. The front end 20F of the sleeve 20 is located forward of the front end 18F of the sheath 18 in the front-rear direction.

(35) If the crimping force by the crimping portion 30 is reduced to avoid a reduction in communication performance, the shielded cable 10 is relatively displaced rearward with respect to the outer conductor 28 and the sleeve 20 when a rearward pulling force is applied to the shielded cable 10. However, at the time of or immediately after the start of a relative displacement of the shielded cable 10, the bent portion 16 on the front end of the folded portion 17 of the braided wire 15 is caught by the front end 20F of the sleeve 20. Thus, a rearward relative displacement of the shielded cable 10 is hindered. Therefore, a reduction in communication performance can be prevented while a connected state of the shield terminal 25 and the shielded cable 10 is maintained.

(36) If the crimping force by the crimping portion 30 is reduced, a fixing force between the sleeve 20 and the sheath 18 possibly becomes larger than that between the crimping portion 30 and the sleeve 20. In this case, when a rearward pulling force is applied to the shielded cable 10, the sleeve 20 is relatively displaced rearward with respect to the crimping portion 30 integrally with the shielded cable 10. However, since the expanded diameter portion 22 having a diameter increasing toward the front is formed on the front end 20F of the sleeve 20, the tapered outer peripheral surface of the expanded diameter portion 22 is caught by the inner peripheral surface of the tapered portion 42 of the crimping portion 30 via the folded portion 17 of the braided wire 15. In this way, a rearward relative displacement of the sleeve 20 with respect to the crimping portion 30 is prevented.

(37) The rear end 22R of the expanded diameter portion 22 is located rearward of the front end 18F of the sheath 18. According to this configuration, when a rearward pulling force is applied to the shielded cable 10, the front end part of the sheath 18 is caught by the inner peripheral surface of the expanded diameter portion 22, whereby a rearward relative displacement of the shielded cable 10 can be prevented.

(38) Since the crimping portion 30 is formed with the biting portions 39 configured to come into contact with and bite into the sheath 18, a rearward relative displacement of the shielded cable 10 with respect to the shield terminal 25 can be prevented by the biting action of the biting portions 39.

(39) The front end 20F of the sleeve 20 is located forward of the front end 18F of the sheath 18.

Thus, with the sheath **18** put on the sleeve **20**, a positional relationship of the sleeve **20** with the sheath **18** in the front-rear direction is unknown. As a countermeasure against that, the front end part of the sleeve **20** is formed with cut portions **23** for exposing the front end **18F** of the sheath **18** to an outer peripheral surface side of the sleeve **20**.

(40) As shown in FIG. 6, the cut portion **23** is in the form of a window hole having an elliptical opening long in the front-rear direction and an opening edge continuous over an entire periphery. Accordingly, the cut portion **23** is not open in the front end **20F** of the sleeve **20**. If the front end **18F** of the sheath **18** is located in an opening region when the cut portion **23** is viewed from an outer peripheral side of the sleeve **20**, a positional relationship of the front end **20F** of the sleeve **20** and the front end **18F** of the sheath **18** can be confirmed through the cut portion **23**. In this way, the sleeve **20** can be arranged at a proper position with respect to the sheath **18**.

Second Embodiment

(41) A second specific embodiment of the present disclosure is described with reference to FIG. 7. A shielded electrically conductive path B of the second embodiment is different in configuration from the first embodiment in a cut portion **50** formed in a sleeve **50**. Since the other configuration is the same as in the first embodiment, the same components are denoted by the same reference signs and the structures, functions and effects thereof are not described.

(42) The cut portion **51** of the second embodiment is an opening elongated in the front-rear direction. The front end of the cut portion **51** is open in a front end **50F** of the sleeve **50**. Accordingly, even if the front end **50F** of the sleeve **50** is arranged at the same position as a front end **18F** of a sheath **18** in the front-rear direction, a positional relationship of the front end **50F** of the sleeve **50** and the front end **18F** of the sheath **18** can be confirmed in the cut portion **51**.

Other Embodiments

(43) The present invention is not limited by the above described and illustrated embodiments, but is represented by claims. The present invention is intended to include all changes in the scope of claims and in the meaning and scope of equivalents and also include the following embodiments.

(44) Although the expanded diameter portion is formed in the front end part of the sleeve in the above first and second embodiments, the sleeve may include no expanded diameter portion.

(45) Although the rear end of the expanded diameter portion is located rearward of the front end of the sheath in the above first and second embodiments, the rear end of the expanded diameter portion may be arranged at a position forward of the front end of the sheath or may be arranged at the same position as the front end of the sheath when the front end of the sleeve is located forward of the front end of the sheath.

(46) Although the crimping portion is formed with the biting portions in the above first and second embodiments, the crimping portion may include no biting portions.

(47) Although the sleeve is formed with the cut portions in the above first and second embodiments, the sleeve may include no cut portions.

(48) Although the sleeve is of an open type to be mounted by winding a member in the form of a flat plate on the shielded cable in the above first and second embodiments, the sleeve may be of a closed type formed into a hollow cylindrical shape.

(49) Although the front end of the sleeve is arranged forward of the front end of the sheath in the above first and second embodiments, the front end of the sleeve may be arranged at the same position as the front end of the sheath in the front-rear direction.

LIST OF REFERENCE NUMERALS

(50) **10** . . . shielded cable **11** . . . coated wire **12** . . . core wire **13** . . . insulation coating **15** . . . braided wire **16** . . . bent portion **17** . . . folded portion **18** . . . sheath **18F** . . . front end of sheath **20** . . . sleeve **20F** . . . front end of sleeve **21** . . . constant diameter portion **22** . . . expanded diameter portion **22F** . . . front end of expanded diameter portion **22R** . . . rear end of expanded diameter portion **23** . . . cut portion **25** . . . shield terminal **26** . . . inner conductor **27** . . . dielectric **28** . . . outer conductor **29** . . . shell body portion **30** . . . crimping portion **31** . . . upper shell **32** . . . lower

shell 33 . . . upper body portion 34 . . . base plate portion 35 . . . first crimp portion 36 . . . second crimp portion 37 . . . first locking portion 38 . . . second locking portion 39 . . . biting portion 40 . . . lower body portion 41 . . . restricting portion 42 . . . tapered portion 50 . . . sleeve 50F . . . front end of sleeve 51 . . . cut portion A . . . shielded electrically conductive path B . . . shielded electrically conductive path

Claims

1. A shielded electrically conductive path, comprising: a shield terminal including an inner conductor and an outer conductor surrounding the inner conductor; a shielded cable including a core wire fixed to a rear end part of the inner conductor, a braided wire surrounding the core wire and a sheath surrounding the braided wire; and a sleeve mounted on an outer periphery of a front end part of the sheath, the braided wire extending from a front end of the sheath being folded rearward and put on an outer periphery of the sleeve and a crimping portion in a rear end part of the outer conductor being crimped to the sleeve while surrounding an outer periphery of a folded portion of the braided wire in a front end part of the shielded cable, and a front end of the sleeve being located at the same position as or forward of the front end of the sheath in a front-rear direction, wherein the front end of the sleeve is located forward of the front end of the sheath, and the sleeve is formed with a cut portion for exposing the front end of the sheath to an outer peripheral surface side of the sleeve.
 2. The shielded electrically conductive path of claim 1, wherein the crimping portion is formed with a biting portion configured to come into contact with and bite into the sheath.
 3. The shielded electrically conductive path of claim 1, wherein an expanded diameter portion having a diameter increasing toward front is formed in a front end part of the sleeve.
 4. The shielded electrically conductive path of claim 3, wherein a rear end of the expanded diameter portion is located rearward of the front end of the sheath.
 5. A shielded electrically conductive path, comprising: a shield terminal including an inner conductor and an outer conductor surrounding the inner conductor; a shielded cable including a core wire fixed to a rear end part of the inner conductor, a braided wire surrounding the core wire and a sheath surrounding the braided wire; and a sleeve mounted on an outer periphery of a front end part of the sheath, the braided wire extending from a front end of the sheath being folded rearward and put on an outer periphery of the sleeve and a crimping portion in a rear end part of the outer conductor being crimped to the sleeve while surrounding an outer periphery of a folded portion of the braided wire in a front end part of the shielded cable, and a front end of the sleeve being located at the same position as or forward of the front end of the sheath in a front-rear direction, wherein the sleeve is formed with a cut portion for exposing the front end of the sheath to an outer peripheral surface side of the sleeve.
 6. The shielded electrically conductive path of claim 5, wherein the crimping portion is formed with a biting portion configured to come into contact with and bite into the sheath.
 7. The shielded electrically conductive path of claim 5, wherein an expanded diameter portion having a diameter increasing toward front is formed in a front end part of the sleeve.
 8. The shielded electrically conductive path of claim 7, wherein a rear end of the expanded diameter portion is located rearward of the front end of the sheath.
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